



Pore structure, internal relative humidity, and fiber orientation of 3D printed concrete with polypropylene fiber and their relation with shrinkage

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ABSTRACT

The shrinkage properties and its mitigation strategies for 3D printed concrete (3DPC) are of great significance, especially for structural applications in construction. This study aims to understand the effect of polypropylene fiber (PP fiber) on the total shrinkage of 3DPC under drying. The rheological properties, mechanical properties, internal relative humidity (RH), pore size distribution and fiber orientation of 3DPC were investigated and compared with cast concrete. Without PP fiber, the total shrinkage of the printed specimens is similar to the cast ones. The addition of PP fiber reduces the total shrinkage for cast specimens more significantly than for printed specimens. This phenomenon is explained from both pore structure and fiber orientation perspectives. A lower total shrinkage is also observed on specimens cut along the printing direction when compared with those cut perpendicularly to the printing direction due to higher quantity of fiber oriented along the printing direction. The total shrinkage of 3DPC is found inversely proportional to the fiber orientation factor.

1. Introduction

The concept of 3D concrete printing was firstly proposed in the 1990s [1,2], and ever since, the concept has been developing very quickly in recent years. Currently, 3D concrete printing based on extruding through nozzle of 3D printer is a commonly accepted technology by researchers and construction industry. The application of this novel fabrication method in construction is reported to reduce the construction wastes by 30%–60%, labour costs by 50%–80%, and construction time by 50%–70% [3,4]. To achieve desired pumping, extruding and printing properties of 3DPC, many studies were carried out, focusing on the rheological properties [5,6], printability [7,8], and their correlation [9]. The hardened properties, including interlayer bond properties [10–12] and mechanical properties, especially the anisotropy of mechanical properties [13–15] of 3DPC have been intensively studied as well. In the design of 3D printable concrete, geopolymer composites [16–18] and mineral admixtures [19–21] have often been added to improve the properties of 3DPC. The properties of 3DPC are not only influenced by mix design parameter but also affected by printing parameter. The relation between printing parameters and the properties of 3DPC has been investigated [22,23].

With the improving understanding of 3D concrete printing, more and more practical applications based on this novel technology

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have been reported, e.g. Striatu bridge [24], Dubai municipality building [25], prefabricated building with 3D printed walls [26] and wind turbine tower segments [27], as shown in Fig. 1.

The cracking risk of concrete has been a major concern of engineers for civil engineering projects. Many factors can influence the cracking of concrete, such as the specimen geometry, external environment conditions (evaporation rate of moisture), concrete composition (strength and elastic modulus) and constraints (boundary conditions or reinforcement) [28]. The shrinkage of concrete is one of the most critical parameters responsible for concrete cracking as it can increase the tensile stress in concrete. When the tensile stress exceeds the tensile strength of concrete, cracking occurs [29]. According to the happening stage, i.e., fresh, hardened, and the origins, e.g. water evaporation, self-desiccation or temperature, different terms are used to denote shrinkage of concrete, such as plastic shrinkage, drying shrinkage, autogenous shrinkage, thermal shrinkage, and carbonization shrinkage [30]. For hardened concrete in drying environment, a combination of autogenous and drying shrinkage is often denoted as the total shrinkage [31]. In this article, we refer drying shrinkage to the time-dependent deformation purely due to water migration from hardened concrete to its ambient environment, autogenous shrinkage refers to the time dependent deformation of hardened concrete due to cement hydration, and total shrinkage refers to the total time dependent deformation of hardened concrete in specific environmental condition.

For 3DPC, the lack of formwork during 3D printing process exposes the freshly deposited concrete filaments to the ambient environment directly. 3DPC is, therefore, more vulnerable to the environment than conventional concrete that is protected from the environment by formwork. Besides, limited by 3D concrete printer and pumping system, the size of aggregates used in 3D printable concrete is quite often below 5 mm, i.e., mortars are often used instead of concrete. The volume fractions of aggregates in 3D printable mortar mixes are 40%–50%, which are much lower than conventional concrete, typically in the range of 60%–75% [32]. Furthermore, lower water to cement (w/c) ratio is often applied to ensure the printability and mechanical properties of concrete, but this often results in higher autogenous shrinkage [33,34]. Some researchers attempted to develop methods for reinforcing 3DPC [35,36]. However, the proposed methods are still far from satisfactory, from the perspective of the interference of the 3D printing process during steel reinforcement layout and the perspective of the strengthening effect of limited steel on the structural performance. Without doubt, the above situations, including relatively high quantities of fine powders, low aggregate, low w/c ratio and lack of steel reinforcement, can bring severe cracking risk to 3DPC [37]. On material basis, it is crucial to quantify the total shrinkage of 3DPC under drying environment, understand its impacting parameters, and study methods to mitigate it.

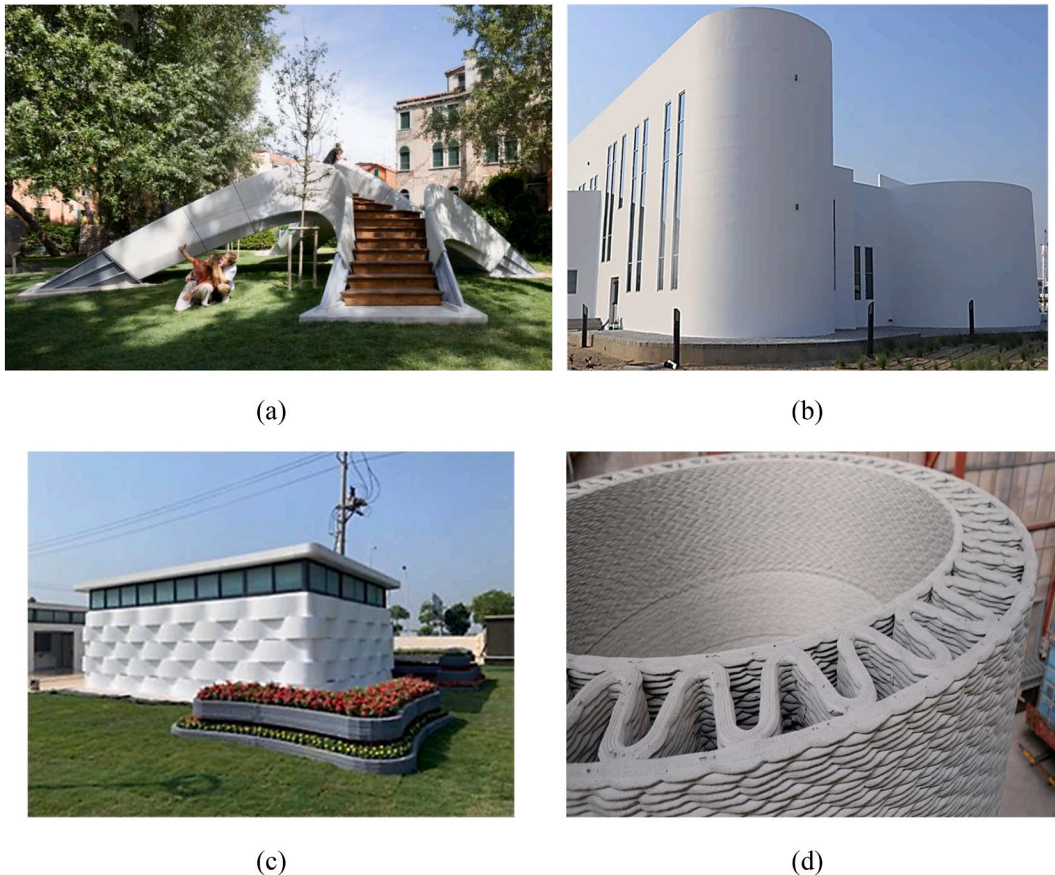


Fig. 1. Practical applications of 3D concrete printing: (a) Striatu bridge; (b) 3D printed Dubai municipality building; (c) Prefabricated building with 3D printed walls; (d) 3D printed wind turbines.

Different research teams have studied the total/drying shrinkage of 3D printable concrete mixes with varying methods on cast and/or printed elements. Le et al. [38] monitored the drying shrinkage of cast 3D printable concrete specimen with the dimension of $75 \times 75 \times 229 \text{ mm}^3$ over six months. The drying shrinkage of the specimen at 20°C and 60% RH is 3–4 times higher than that cured at 20°C and 100% RH. Slavcheva et al. [39] measured the total shrinkage of cast specimens (3D printable concrete) with $10 \times 40 \times 160 \text{ mm}^3$ in size under various relative humidity from 20%, 30%, 50% to 70%. The results showed that diminishing the RH from 70% to 20% increases the total shrinkage by 2.3 times. Gerrit M. Moelich et al. [40] used the digital image correlation (DIC) technique to calculate the displacement of printed elements. The results showed that the plastic shrinkage mainly occurs within the first 2 h after printing, and the strain rate is several times higher than the traditional concrete. They also found that the higher surface-to-volume ratio in thin 3DPC elements increases the pore moisture loss and the strain development. J. Van Der Putten et al. [41] applied DIC to evaluate the shrinkage of 3DPC, starting from 10 min after mixing up to 6 days. The deformations both in longitudinal and transverse directions of single filament are similar. Yu Zhang et al. [42] measured the drying shrinkage of 3D printed specimens in dimension of $100 \times 100 \times 400 \text{ mm}^3$ at $20 \pm 2^\circ\text{C}$ and $50 \pm 5\%$ RH and found that the shrinkage reaches 807–840 μe after 70 d for printed specimens.

Fibers are often added to concrete to improve its properties. Fibers with high elastic modulus, e.g. basalt fiber, steel fiber, carbon fiber, can increase the rigidity, tensile and shear strength of concrete [43,44]. On the other hand, fibers with low elastic modulus, e.g. polypropylene (PP) fiber, polyvinyl alcohol fiber, have good ductility and are often used to enhance the crack resistance of cement-based materials [45]. So far, various types of fibers, such as PP fiber [46,47], steel fiber [48], polyvinyl alcohol fiber [13], basalt fiber, and carbon fiber [49], have been utilized in 3DPC. Among these fibers, PP fiber is widely used due to its relatively low cost, high fire resistance [21,50], and high effectiveness in plastic shrinkage mitigation and in cracking control which is the focus of this study [8]. Behzad Nematollahi [51] investigated the effect of PP fibers with a length of 6 mm and volume fraction from 0.25% to 1.00% on both fresh and hardened properties of 3D printed mortar with a maximum sand size of 465 μm . The results show that the shape-retention capability of printed mortar is improved with the increase of fiber dosage. Mien V. Tran [52] studied the effect of PP fiber dosage (1.35–5.4 kg/m^3) on the shrinkage of 3DPC in 24 h after mixing, but no significant influence of PP fiber on the shrinkage in the time span of testing.

This study focuses on the effect of PP fiber on the properties of 3DPC under drying condition. The experimental investigation includes rheological, mechanical properties, shrinkage, internal RH, mass loss, pore size distribution, and fiber orientation for 3DPC with PP fibers. The total shrinkage of printed specimens cut horizontally and perpendicularly along the printing direction is assessed and compared with that of cast specimens. The relationship among total shrinkage, pore distribution and fiber orientation are discussed.

2. Materials and experiment

2.1. Raw materials and mix proportions

The 3DPCs with maximum sand size of 1.18 mm and PP fiber length of 6 mm were prepared. Table 1 presents the chemical compositions of Portland cement (P-II 52.5) and silica fume used in this study. Viscosity modifying agent (VMA) with a viscosity of 6000 mPa s was used to increase the viscosity of 3DPC. Polycarboxylate superplasticizer (PCE) was adopted to adjust the workability of printable concrete.

The properties of PP fibers used are given in Table 2.

To avoid the blockage of the printer nozzle by 3DPC, relatively low volume dosages of PP fiber, 0, 0.1%, 0.3% and 0.5%, were considered. With such dosages, PP fibers can play a role in reducing the shrinkage of concrete [53]. The mix proportion of the 3D printable mixture is listed in Table 3.

For the preparation of 3DPC, dry materials, i.e., cement, silica fume, sand, PCE, and VMA were mixed in a Non-Tilting Drum Mixer for 1 min at a speed of 48 rpm. Water was then gradually added and mixed for another 4 min during which the manual dispersed PP fibers were added. The total mixing time was 5 min to ensure all ingredients were thoroughly mixed. The cast specimens were prepared following the same mixing procedure as 3DPC. 2-Minutes vibration was applied to cast specimens to facilitate molding.

2.2. Printing process

A $3 \times 3 \times 3 \text{ m}^3$ gantry-type 3D concrete printer manufactured by Nanjing Jiayi Digital Additive Technology Research Institute was used to print (Fig. 2 (a)). Material was extruded through a 20 mm circular nozzle equipped with a printer. The extrusion speed was 70 rpm, the nozzle linear speed was 150 mm/s, and the printing time between two layers was 13 s. The height of each printed filament was around 12 mm, and the width was around 25 mm (Fig. 2 (b)).

Immediately after deposition, a rectangular PVC box ($2200 \times 1200 \times 500 \text{ mm}^3$) was used to cover the printed elements. Inside the box, saturated NaBr solution and silica gel was placed in respective containers to help stabilizing the relative humidity inside the box

Table 1
Chemical compositions of cement and silica fume by XRF (%).

	CaO	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	MgO	SO ₃	K ₂ O	Na ₂ O	P ₂ O ₅	TiO ₂	LOI
PC	64.55	17.47	3.82	2.96	1.00	3.46	0.65	0.26	0.10	0.21	2.40
SF	–	93.10	0.26	0.06	0.14	0.57	0.22	0.34	0.05	0.01	1.68

Table 2
Properties of PP fiber.

Length (mm)	Diameter (μm)	Density (g/cm^3)	Elastic modulus (MPa)
6	100 ± 10	0.91	3500

Table 3
Mix proportion of concrete for 3D printing (kg/m^3).

Notation	Cement	SF	Sand	Water	PCE	VMA	Fiber
F0.0%	697.4	60.6	1212.9	227.4	1.29	0.38	0.0
F0.1%	697.4	60.6	1212.9	227.4	1.29	0.38	0.91
F0.3%	697.4	60.6	1212.9	227.4	1.29	0.38	2.73
F0.5%	697.4	60.6	1212.9	227.4	1.29	0.38	4.55



Fig. 2. (a) Concrete 3D printer; (b) printed concrete.



Fig. 3. Brookfield rheometer with a vane spindle of $20 \text{ mm} \times 10 \text{ mm}$ in size.

around $60 \pm 5\%$. The specimens were cut from the printed elements 2 h after adding water. After the final setting time of the material (about 5 h), the specimens were moved to the constant temperature and humidity chamber ($RH = 60 \pm 2\%$, $T = 20 \pm 1^\circ\text{C}$) until being tested.

2.3. Rheology tests

Specific rheological properties are needed to ensure the printability of 3DPC [5]. The rheological properties of 3DPC with different fiber dosages were tested. Brookfield rheometer (Fig. 3) with a vane spindle of $20\text{ mm} \times 10\text{ mm}$ in size within a beaker of 500 ml was used to conduct rheological tests. Static yield stress, dynamic yield stress, and plastic viscosity were measured.

A constant shearing rate of 0.1 s^{-1} was kept for 3 min to measure the static yield stress [54]. The shear stress peak was measured as the static yield stress [8]. The same testing protocol was applied at 10, 20, 30 and 40 min after adding water to dry powder, i.e., 5, 15, 25, and 35 min after the mixing procedure. The following protocol was applied to measure dynamic stress and viscosity: linearly increase the rotation of vane spindle from 0 s^{-1} to 50 s^{-1} within 60 s; then decrease the rotation to 0 s^{-1} linearly in the following 60 s [42]. Bingham equation, $\tau = \tau_0 + \eta\dot{\gamma}$, was applied to fit the curve with decreasing shear rate from 50 s^{-1} to 0 s^{-1} , where τ is shear stress, τ_0 is dynamic yield stress, η is plastic viscosity and $\dot{\gamma}$ is shear rate [55].

2.4. Mechanical properties

To clarify the influence of fibers on the mechanical properties for 3DPC, specimens cut from printed elements were loaded in X, Y, and Z directions, as shown in Fig. 4. The flexural and compressive strength tests were carried out according to Chinese National Standard GB/T 17,671–1999 [56]. The dimension of specimens was $40 \times 40 \times 160\text{ mm}^3$ for flexural test and $40 \times 40 \times 40\text{ mm}^3$ for compressive test, respectively. Three parallel specimens were prepared for flexural test and six for compressive test. Before the test, specimens were cured at $20 \pm 1^\circ\text{C}$ and $60 \pm 2\%$ RH. Mechanical tests were conducted at 1 d, 7 d and 28 d.

The specimens for flexural and compression tests were cut from printed elements, either horizontally or perpendicularly, to explore the effect of the interlayers on 3DPC. As presented in Fig. 4, the specimens cut horizontally to the printing direction, two loading directions, X and Z, were applied. The specimens cut perpendicularly to the printing direction were loaded in Y direction. The cast specimens were prepared for comparison with printed specimens.

2.5. The total shrinkage

The specimens ($40 \times 40 \times 160\text{ mm}^3$) were cut from the printed elements 2 h after mixing, as shown in Fig. 5. Digital length comparators with an accuracy of 0.001 mm were used to measure the length of specimens. Copper heads were inserted into two ends of the specimens immediately after the specimens were cut from printed elements. To evaluate the effect of interlayers on shrinkage properties of the printed specimens, the shrinkage was tested in two directions: along the printing direction (1 interlayer in parallel with the direction of measurement) and perpendicular to the printing direction (6 interlayers perpendicular to the direction of measurement), as presented in Fig. 5(a–c). The cast specimens with the same dimension were prepared for comparison. The specimens were cured at $60 \pm 5\%$ RH, $20 \pm 2^\circ\text{C}$ after printing, and moved into an environmental container with stable temperature and relative humidity ($RH = 60 \pm 2\%$, $T = 20 \pm 1^\circ\text{C}$) after the final setting time (5 h). The initial length measurement started at 24 h after the addition of water, then measured every day in the first week, and at 10 d, 14 d and 28 d.

2.6. Internal relative humidity

As shown in Fig. 6 (a), relative humidity sensors were used to measure the internal RH evolvement of 3DPC. The evolution of relative humidity at various depths was measured for the printed specimens that contain one vertical interface and three horizontal

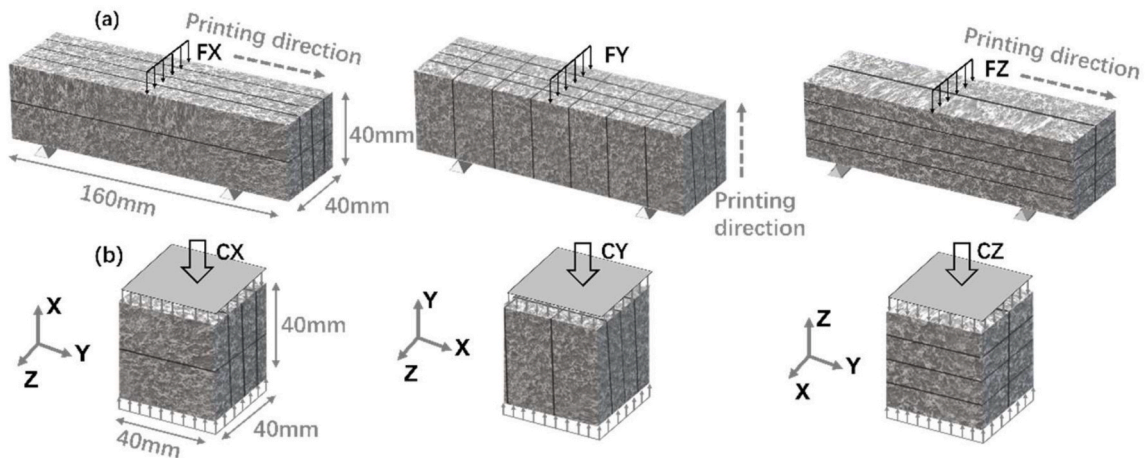


Fig. 4. Loading of specimens: (a) flexural; (b) compressive test.

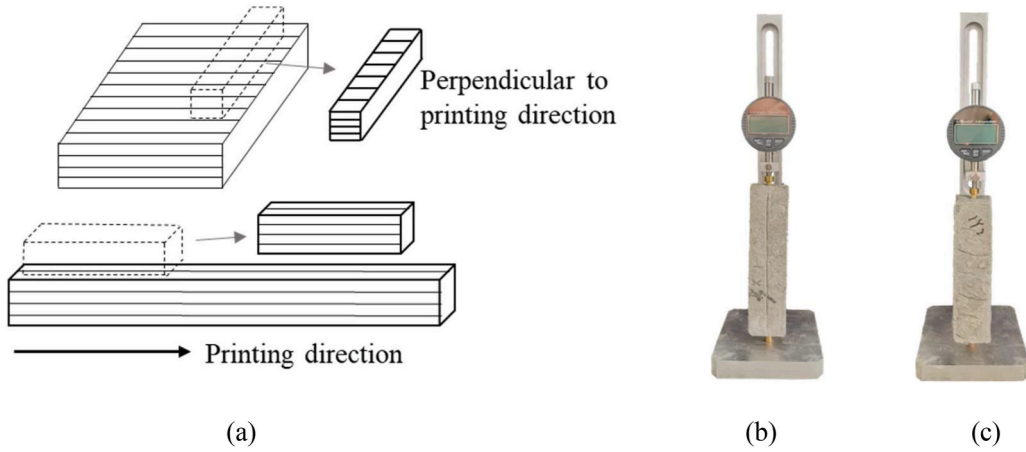


Fig. 5. Shrinkage measurement: (a) cutting specimens from 3D printed elements; (b) length test along printing direction; (c) length test perpendicular to printing direction.

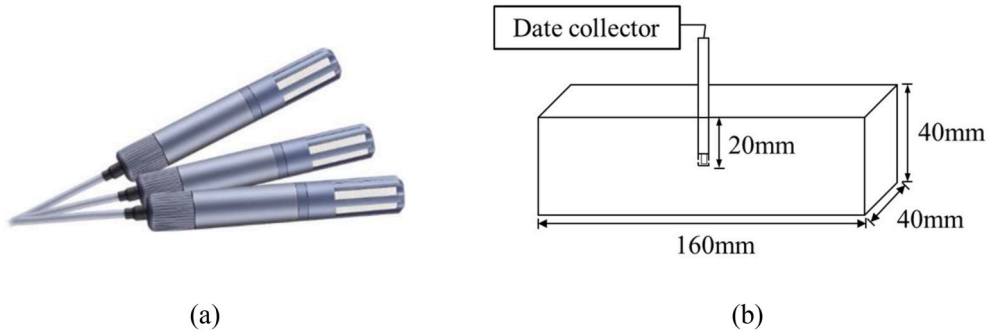


Fig. 6. Internal RH measurement: (a) RH sensors; (b) Location of sensor in concrete specimen.

interfaces. The sensors TP2307, with an accuracy of $\pm 1.8\%$, were manufactured by Shenzhen Toprie Electronics Co., Ltd. The sensor is equipped with a humidity-dependent resistor. When the relative humidity varies, the water vapor goes through the casing (the sensor does not contact the concrete) and is adsorbed on the resistor. This process leads to the variation of dielectric constant of the humidity-dependent resistor, which is proportional to the relative humidity, and the relative humidity can thus be measured. The RH sensors were located in the center of specimens with $40 \times 40 \times 160 \text{ mm}^3$ in size, as depicted in Fig. 6 (b). The initial measurement began just after deposition. Saturated NaBr solutions were applied to maintain the curing condition at $60 \pm 5\%$ RH under $20 \pm 2^\circ\text{C}$.

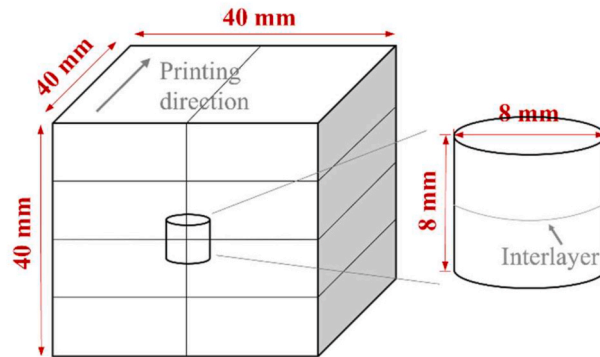


Fig. 7. Specimen cut from 3D printed element with interlayer for nano-CT scan.

2.7. Mass loss

To estimate the mass change of printed specimens, the weight (accuracy of ± 0.01 g) of specimens under drying condition ($T = 20 \pm 1$ °C and $RH = 60 \pm 2\%$) were firstly measured at 24 h, every day in the first week, and then measured at 10 d, 14 d and 28 d. The printed specimen with one vertical interface and three horizontal interfaces in the size of $40 \times 40 \times 160$ mm³ was tested. At least three specimens were tested for each group.

2.8. MIP to determine pore structure

The porosity of concrete was determined using mercury intrusion porosimetry (MIP). The porosity was calculated as the ratio of the total pore volume with pore diameter larger than 5.5 nm to the volume of the specimen. Specimens with the diameter of 3–6 mm were extracted from the center of specimens with the dimension of $40 \times 40 \times 40$ mm³, which were cured in a drying room at 20 ± 1 °C and $60 \pm 2\%$ RH for 28 d. For each sample, critical pore size was defined as the size of pore with maximum volume fraction in the derivative of the cumulative curve. The pore size distribution was obtained by averaging two repetitive specimens.

2.9. X-ray nano computed tomography to characterize fiber orientation

A Zeiss Xradia 510 Versa X-ray nano computed tomography (nano-CT) was adopted to scan the specimens cut from 3D printed elements cured for 28 d, as shown in Fig. 7. Cylinder specimens with height and diameter of 8 mm were used for nano-CT scanning. The scanning time was 2 s per image, and the entire scanning process took 1 h for each specimen. For each specimen, about 800 images were taken. The resolution of the images was 1000×1024 pixels with a pixel size of 9.67 μ m. The Dragonfly software [57] was used to analyze fiber orientation in 3D reconstructed specimens.

As the greyscale of PP fibers is often similar to that of pores in images, it is difficult to identify fiber phase from the image by the threshold segmentation method, as shown in Fig. 8(a and b). Therefore, deep learning (DL) method was applied to analyze the fiber distribution. Deep learning is a method for visual classification involved in Convolutional Neural Network (CNN) technique [58]. For classification, nine images were delineated by grey level and pattern of fiber phase manually. The corresponding segmentation maps train the network until its outputs match the desired objection. As shown in Fig. 8 (c), the fibers are marked in blue.

The cast specimen with 0.5% PP fiber was used as a criterion specimen. The dimension of the criterion specimen was the same as the specimens tested for nano-CT, as shown in Fig. 7. After being trained, the error criterion of the model was 4%, and the trained network was then ready to apply to all images.

From the 3D reconstruction of fiber orientation for nano-CT images, the main fiber direction was defined by an angle (φ), the angle between fiber direction and Z coordinate (range from 0 to 90°), as shown in Fig. 9.

To quantify the fiber distribution, fiber orientation factor was defined as η_φ (Eq. (1)), [59]. Here, the fiber orientation factor η_φ can be calculated as an average fiber orientation toward the Z coordinate. The closer the value of fiber orientation factor gets to 1, the more pronounced the fiber alignment. For the perfectly well-distributed fiber, η_φ is 0.5. For cast sample, η_φ is in the range of 0.6–0.9 [60].

$$\eta_\varphi = \frac{1}{N_T} \sum_{i=1}^{N_T} \cos(90 - \varphi_i) \quad (1)$$

Where N_T is the total number of fibers within specimens.

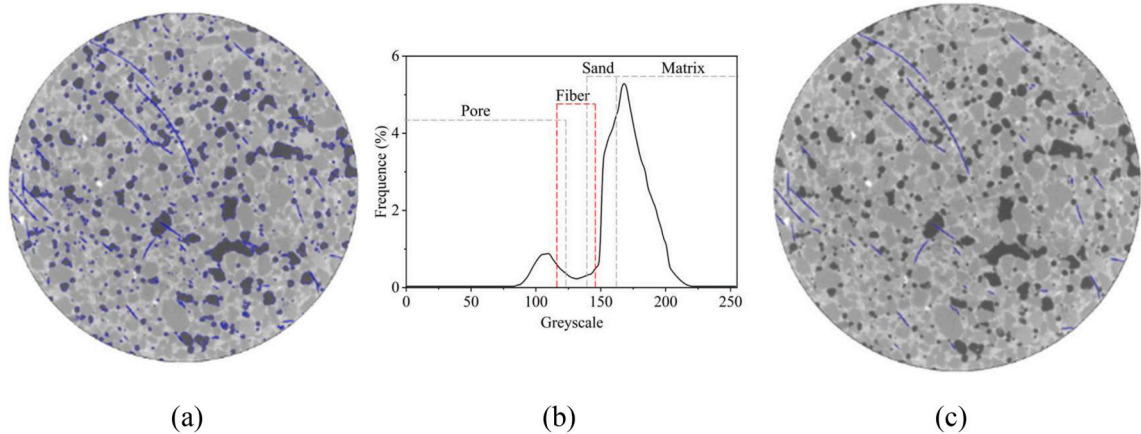


Fig. 8. The results of fiber extraction: (a) threshold segmentation method; (b) greyscale of PP fibers overlap partially with that of pores; (c) deep learning result.

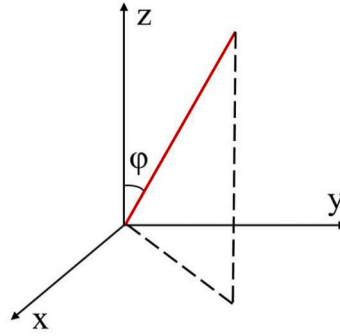


Fig. 9. Illustration of PP fiber geometrical parameter ϕ .

3. Results and discussion

3.1. Rheological properties

3.1.1. Static yield stress

Shear stress in the function of fiber dosage for fresh materials is plotted in Fig. 10 (a). The maximum shear stress refers to the static yield stress. The initial static yield stress is obtained at 10 min after adding water, or 5 min after the mixing procedure. Adding fiber in general increased the static yield stress compared with the material without fiber (Fig. 10 (b)). All the static yield stresses with different PP fiber dosages were printable, i.e., within the range of 1.5–2.5 kPa [8].

Fig. 11 presents the static yield stress evolution from 10 min to 40 min after water addition. The static yield stress increased linearly with time for all considered fiber dosages (0–0.5%). This trend is in agreement with the research of N. Roussel [61]. When fitting the static yield stress evolution versus time with the formula of N. Roussel [62]: $\tau_0(t) = \tau_0 + A_{\text{thix}}t$, where $\tau_0(t)$ is the static shear stress at time t , τ_0 is the initial static yield stress, A_{thix} is the structuration rate due to thixotropy, it can be found that the structuration rate (growth rate of static yield stress) increases with fiber dosage.

3.1.2. Dynamic yield stress and plastic viscosity

The extrudability of 3DPC is mainly affected by the dynamic yield stress [63]. The dynamic yield stress of 3DPC with PP fiber obtained at 10 min after the addition of water is shown in Fig. 12 (a). Compared with the mixture without PP fiber, the mixtures containing PP fiber recovered to flow slower due to the twining of fiber [64].

The plastic viscosity and dynamic yield stress with different dosages of PP fiber are obtained by fitting the descending lines based on the Bingham model, as shown in Fig. 12 (b). The evolution of dynamic yield stress and plastic viscosity with fiber dosage is shown in Fig. 12 (c). With the increase of PP fiber dosage, the plastic viscosity increased gradually from 4.8 to 8.7 Pa s, which has a similar trend as the static yield stress shown in Fig. 10 (b). The dynamic yield stress increased from 159.1 to 214.7 Pa. A similar result can be found in the research of Y. Weng [65].

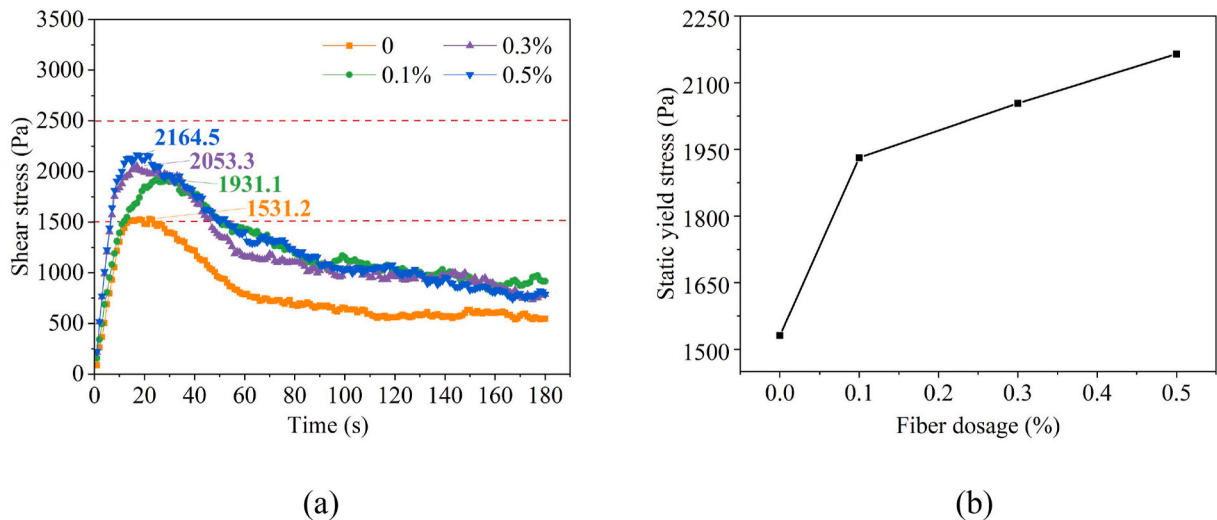


Fig. 10. Static yield stress: (a) shear stress curves; (b) static yield stress versus fiber dosage.

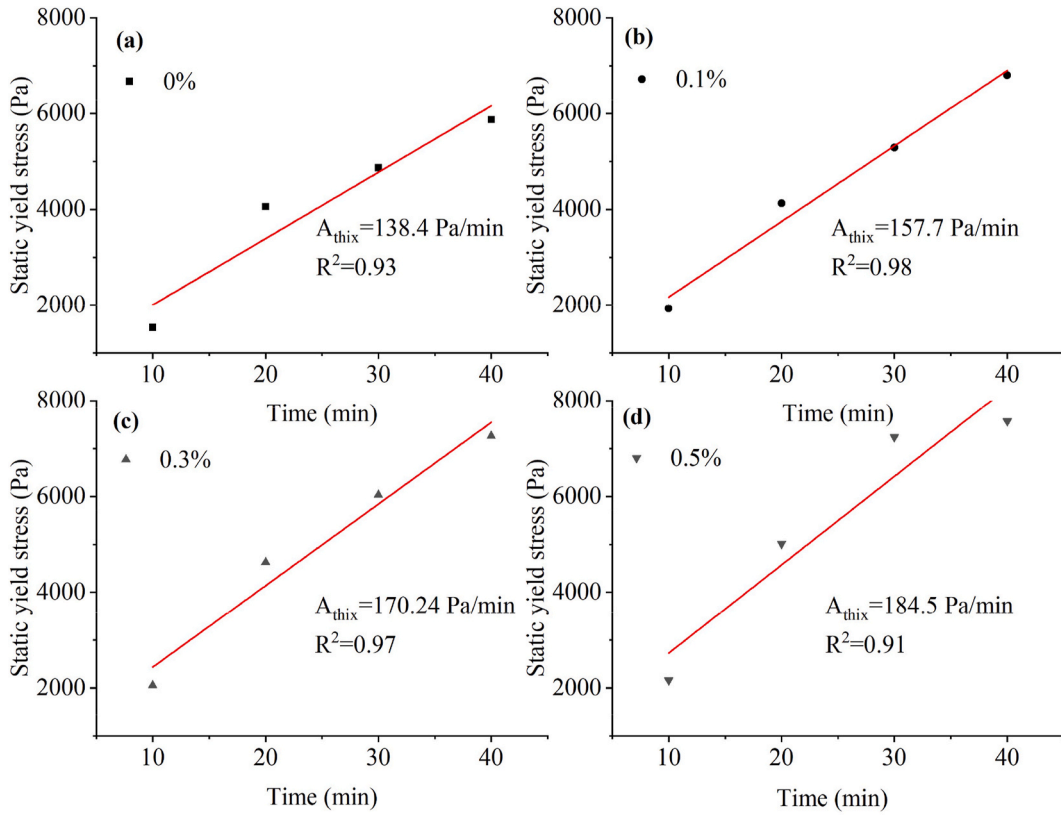


Fig. 11. Static yield stress evolution as a function of time after the addition of water.

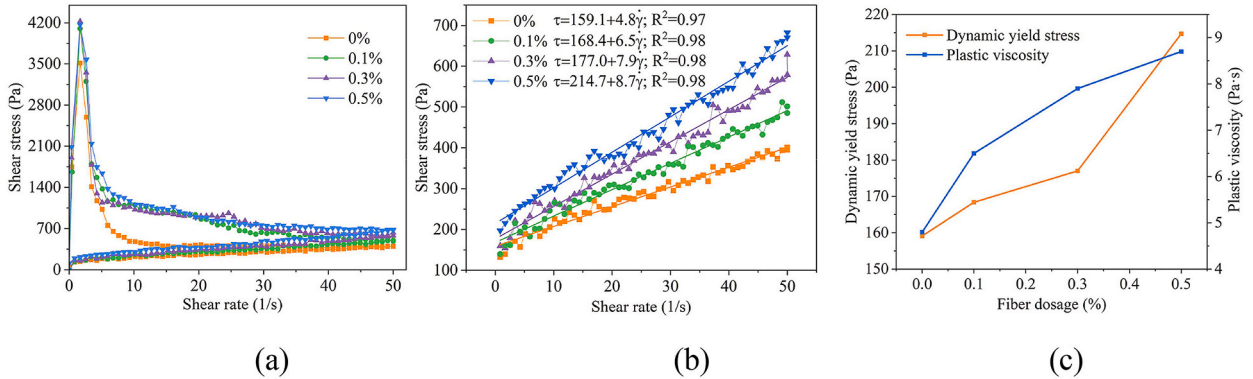


Fig. 12. Dynamic yield stress: (a) effect of fiber dosage; (b) fitted Bingham model; (c) dynamic yield stress and plastic viscosity in function of fiber dosage.

3.2. Mechanical properties

The flexural strength of printed and cast specimens is shown in Fig. 13. For printed specimens, the specimen with 0.3% PP fiber had the highest strength compared with other fiber dosages, especially at 28 days. A significant drop of flexural strength was observed for specimen with fiber dosage of 0.5% compared with 0.3%. For cast specimens, flexural strength did not seem to be influenced by the dosage of PP fiber.

When comparing the flexural strength obtained in different loading directions on 3DPC, the strength loaded in Y direction was lower than the other two directions (X, Z). The addition of PP fiber kept the similar tendency. The main reasons will be discussed from the porosity analysis and the fiber orientation in sections 3.5 and 3.6.

From Fig. 14, it can be seen that the influence of PP fiber on compressive strength for printed and cast specimens was marginal. Concrete with the addition of 0.3% fiber seems to have the highest compressive strength in comparison to other fiber dosages. It can be

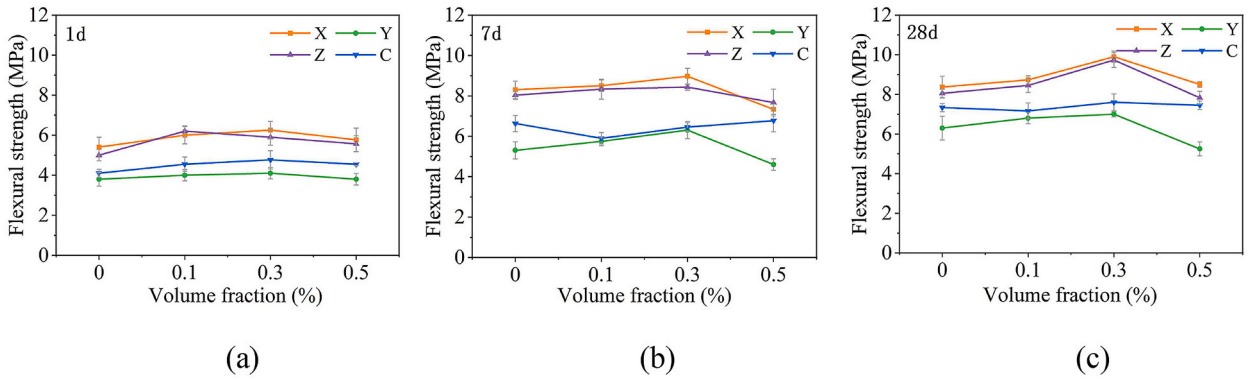


Fig. 13. Effect of fiber dosage on flexural strength: (a) 1 d; (b) 7 d; (c) 28 d. Here, X, Y, Z: printed specimen loaded in X, Y, Z directions; C: cast specimen, as shown in Fig. 3.

also observed that the compressive strength of 3DPC in three different directions was similar, inducing less anisotropy on compressive strength when compared with that of flexural strength. These results agree with those obtained in previous study [66].

3.3. Total shrinkage

The total shrinkage deformation of printed and cast specimens as a function of age is shown in Fig. 15. For each mixture, three

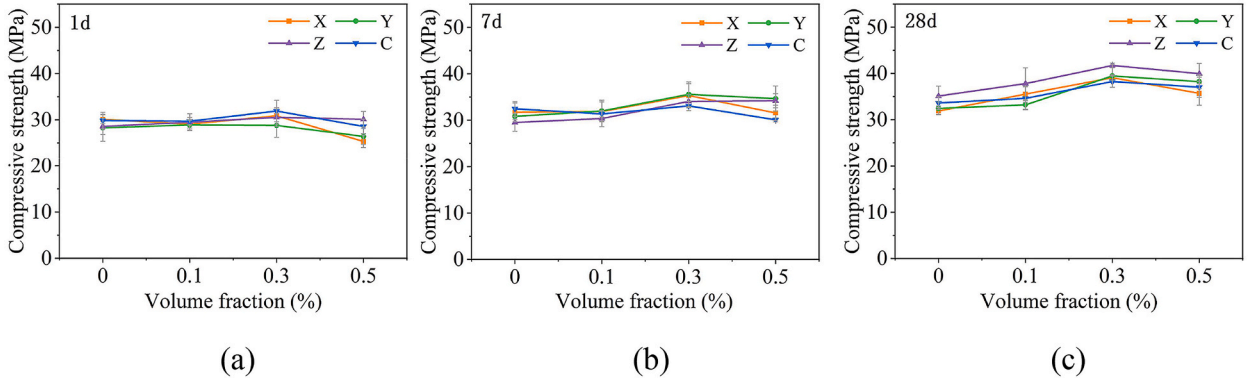


Fig. 14. Effect of fiber dosage on compressive strength: (a) 1 d; (b) 7 d; (c) 28 d. Here, X, Y, Z: printed specimen loaded in X, Y, Z directions; C: cast specimen, as shown in Fig. 3.

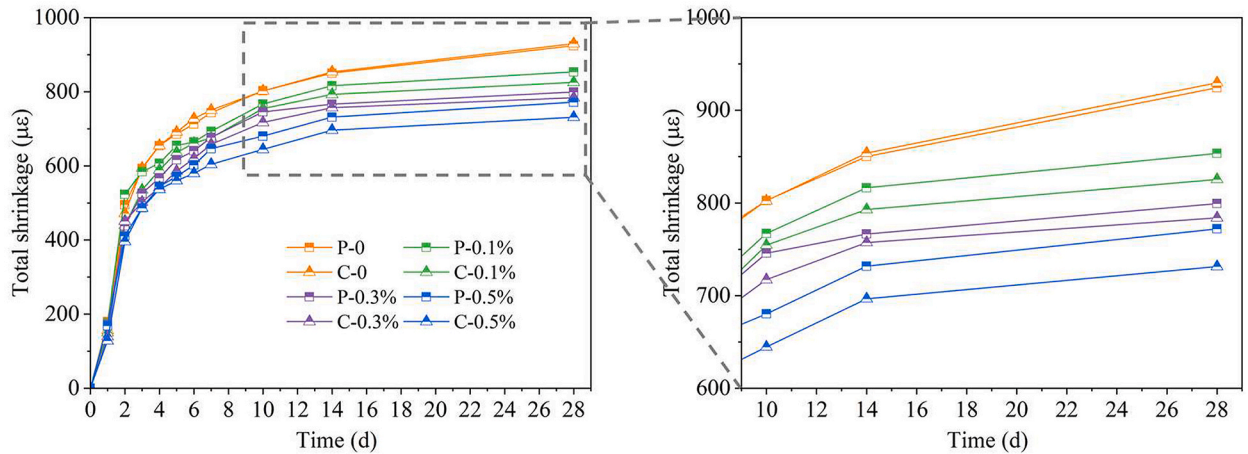


Fig. 15. Comparison of shrinkage between printed and cast specimens with different fiber dosage. P: the average shrinkage of printed specimens (PX and PY); C: cast specimen.

specimens were tested and the average shrinkage deformation was calculated at given time. The variation coefficient of the average shrinkage varied from 2% to 13%. From Fig. 15, for given time, the total shrinkage was decreased with increasing fiber dosages from 0 to 0.5%. For specimen without fiber, the total shrinkage of printed specimens (the average of printed specimens measured along the printing direction and perpendicular to the printing direction) was close to that of the cast specimens, as shown in Fig. 15. However, when PP fiber was added, the shrinkage of printed specimens was higher than the cast ones.

Quantitatively, we introduce shrinkage reduction rate (η_t), defined as $\eta_t = (\varepsilon_{r,t} - \varepsilon_{f,t}) / \varepsilon_{r,t} \times 100$, where $\varepsilon_{r,t}$ is the shrinkage of reference specimen without PP fiber at age of t , $\varepsilon_{f,t}$ is the shrinkage of specimen with PP fiber at age of t . Table 4 summarizes the reduction rate η_t at various ages. The shrinkage reduction rate increased by the increased PP fiber dosage, owing to the inhibiting effect of PP fiber on the shrinkage of 3DPC. The addition of 0.5% PP fiber for printed specimen induced 19.6% reduction. Furthermore, the reduction rate of printed specimens was lower than the cast specimens. The results indicate that the effect of PP fiber on shrinkage of printed concrete is influenced by printing process.

From Fig. 16, it can be noticed that the total shrinkage of PY-0 (printed specimen measured perpendicularly to printed specimen) was slightly higher than that of PX-0 (printed specimen measured along the printed direction). It seems that interfaces may impact on shrinkage to some extent. With respect to the printed specimens with the PP fiber (0.1%–0.5%), the total shrinkage of PY was relatively higher than that of PX, which could be attributed to the fiber alignment along the printing direction for PX specimens.

3.4. Internal RH and mass loss

The internal RH evolution of the printed specimens with fiber dosages from 0 to 0.5% is shown in Fig. 17 (a). The results showed that the addition of PP fiber resulted in higher internal relative humidity at given time, and the internal RH evolvement of specimens with 0.1% and 0.3% PP fiber was similar and the highest, while that with 0.5% PP fiber was lower, though it was a little higher than that without fibers. The maximum difference of internal RH between 0.1% and 0.5% PP fiber is 2.6%, whereas the accuracy of sensors is 3.6% for two measurements. Hence, the difference is not significant. But it was found that all the internal RH of specimens with PP fiber were higher than that without PP fiber, the result indicates that the addition of PP fiber can inhibit moisture evaporation.

The mass loss of printed specimens with the PP fiber dosages of 0–0.5% is shown in Fig. 17 (b). The mass loss increased to around

Table 4

The shrinkage reduction rate of specimens with fibers compared with the specimens without fiber (%) at various ages.

Age	Fabrication method	0.1% PP fiber	0.3% PP fiber	0.5% PP fiber
1 d	Print	2.4 ± 0.16	4.3 ± 0.18	6.4 ± 0.18
	Cast	4.4 ± 0.36	10.9 ± 0.31	18.2 ± 0.39
3 d	Print	1.5 ± 0.06	11.3 ± 0.06	17.9 ± 0.06
	Cast	9.9 ± 0.12	15.5 ± 0.13	18.2 ± 0.13
7 d	Print	6.8 ± 0.06	9.0 ± 0.06	13.1 ± 0.06
	Cast	10.0 ± 0.13	12.3 ± 0.12	19.5 ± 0.12
14 d	Print	4.0 ± 0.06	9.8 ± 0.06	13.9 ± 0.06
	Cast	7.3 ± 0.12	11.3 ± 0.12	16.4 ± 0.12
28 d	Print	6.5 ± 0.06	12.4 ± 0.06	15.4 ± 0.06
	Cast	9.2 ± 0.12	13.8 ± 0.12	19.6 ± 0.12

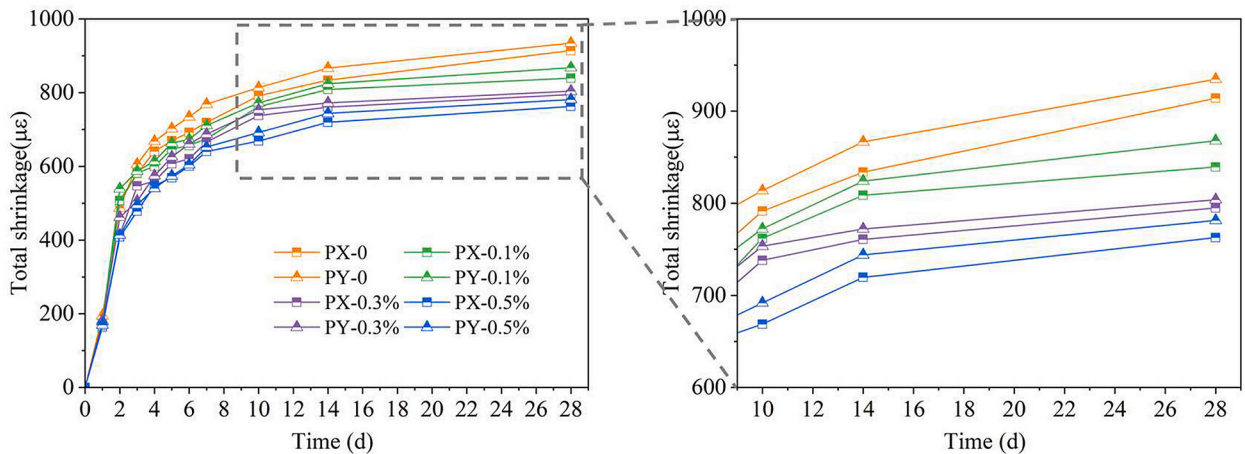


Fig. 16. Comparison of shrinkage of printed specimens along different directions with different fiber dosage. PX: printed specimen measured along the printed direction; PY: printed specimen measured perpendicularly to printed specimen.

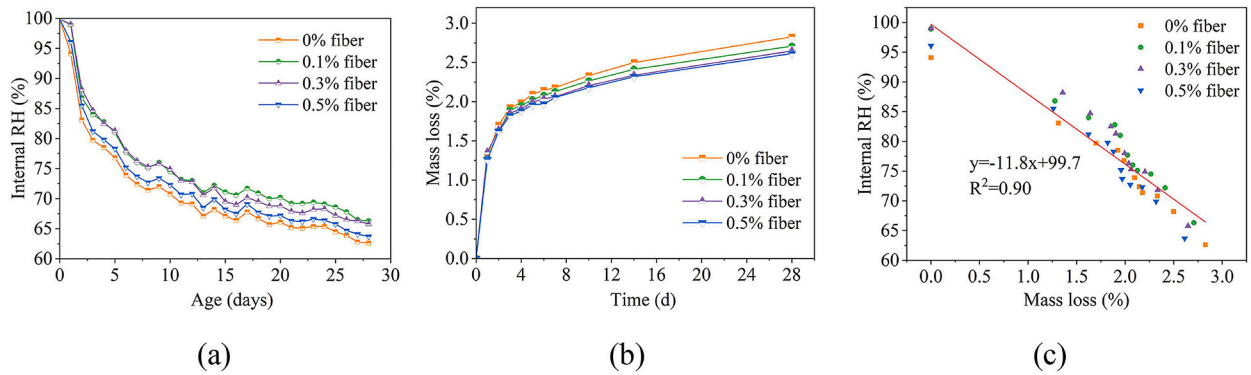


Fig. 17. Evolution of internal RH and mass loss with the fiber volume fraction of 0–0.5%: (a) internal RH; (b) mass loss; (c) the correlation between internal RH and mass loss.

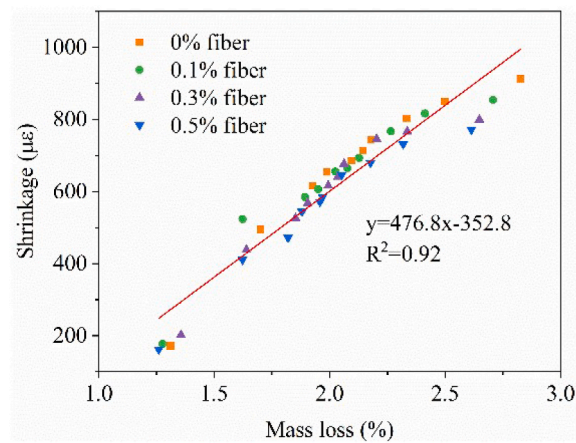


Fig. 18. The correlation between shrinkage and mass loss.

2.2% during the first 10 d and kept increasing up to around 2.5% at 28 d. The addition of PP fiber eased the mass loss compared with the reference group. The correlation between the internal RH and mass loss is illustrated in Fig. 17 (c). It was obvious that the internal RH of concrete exhibited a good linear inverse correlation with the mass loss.

The correlation between shrinkage and mass loss is shown in Fig. 18. The total shrinkage estimated in this study was proportional to mass loss, which is similar to the trend observed by Xianwei Ma [67]. Mass loss of specimen is determined by the evaporation of moisture in drying condition. The ongoing moisture loss induces the capillary pressure and the shrinkage strain increases [40].

3.5. Pore size distribution

Fig. 19 presents the porosity as a function of fiber dosage for printed and cast specimens cured in drying condition at 28 d. The porosity of the printed specimen was generally lower than cast specimen. The porosity decreased with the PP fiber dosage increased from 0% to 0.1%. However, the porosity increased when the fiber dosage was up to 0.5%. This phenomenon may be related to the rheological properties in the fresh stage: the addition of 0.5% PP fiber increases the viscosity significantly, the extrusion process for printed elements or vibration process for cast specimens cannot densify the concrete and the porosity increases compared with the specimens with low fiber dosage [52].

In Fig. 20, the pore distribution of printed and cast specimens is compared. Three typical peaks can be identified: the first peak in the range of 5.5 nm–50 nm belonging to mesopores, the second peak in the range of 50 nm–10 μm responding to macropores, and the third peak above 10 μm corresponding to voids/microcracks [68]. Previous studies found that pores below 50 nm play an important role in shrinkage, whereas pores in 50 nm–10 μm and air voids above 10 μm can affect mechanical properties [69].

From Fig. 20, the critical pore size of samples with the PP fiber of 0–0.5% by volume fraction was located at 50 nm–10 μm (macropores). The critical pore size of the printed specimen was larger than the cast specimen for the group without fiber. However, incorporating PP fiber considerably decreased the number of macropores, especially for the printed specimens. In addition, the pore size peak of the printed specimen below 50 nm was shifted from 62.5 nm to 32.4 nm compared with the cast specimen. But for the 0.5% PP fiber dosage group, the apparent peak shift was not observed, possibly due to the tangling or twining of soft PP fibers with higher dosage [64].

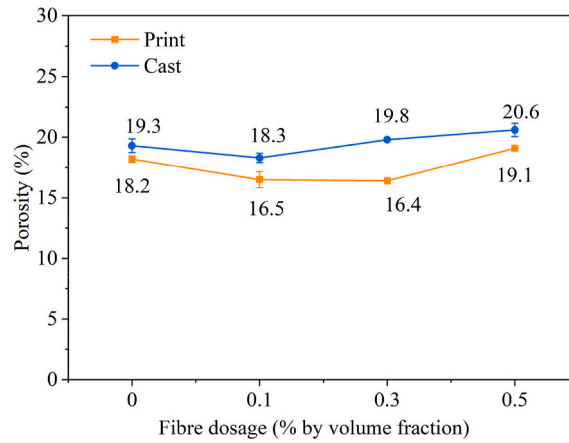


Fig. 19. Porosity of concretes with different dosages of PP fiber.

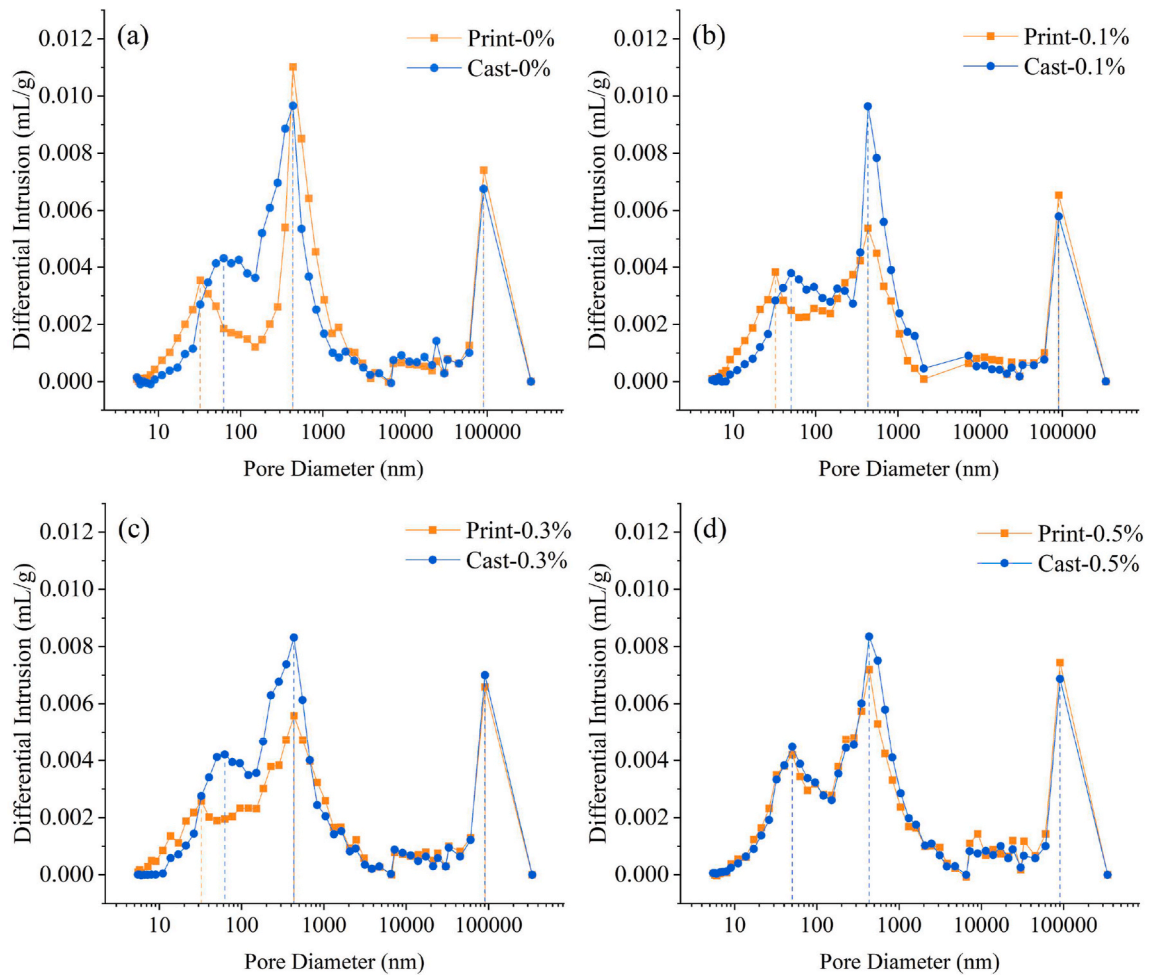


Fig. 20. Pore size distribution of printed and cast specimens with different dosages of PP fiber.

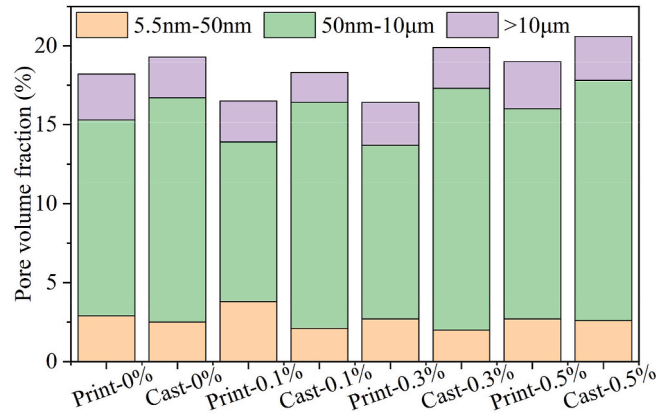


Fig. 21. Volume fraction of different pore ranges for printed and cast specimens with PP fibers.

Fig. 21 Further illustrates the pore volume fraction in the range of 5.5 nm–50 nm, 50 nm–10 μm and >10 μm, respectively. The largest fraction of pores for all the specimens fell into 50 nm–10 μm. Printed specimens occupied a higher volume fraction of pores exceeding 10 μm than cast specimens. Specifically, the pore volume fraction in the range of 5.5 nm–50 nm of printed specimens was close to that of the cast specimens without PP fiber. However, when PP fiber was added, the pore volume fraction of printed specimens in this range was higher than the cast ones. P.K. Mehta et al. [70] concluded that the development of total shrinkage is mainly related to the pore size smaller than 50 nm. The more the pore size smaller than 50 nm, the higher the total shrinkage. The observed higher pore volume fraction of pores in the range of 5.5 nm–50 nm for printed specimens with PP fiber can explain the higher shrinkage compared with cast specimens (Fig. 15).

3.6. Fiber orientation

To understand the difference in mechanical properties (Fig. 13), shrinkage between printed and cast specimens (Fig. 15), and shrinkage between PX and PY of printed specimens (Fig. 15), PP fiber orientation in concrete was characterized by Nano-CT. The fiber distributions in reconstructed samples with identified fiber (in green) are presented in Fig. 22.

It can be seen from Fig. 22 that fibers in the printed specimens oriented in parallel to the printing direction. For the cast specimens, the fibers were more randomly distributed. For conventional concrete, the distribution of fiber in concrete is affected by fiber type, volume dosage of fiber, mix design, mix procedures, and the wall effect caused by the formwork [71]. However, for the 3DPC, the extrusion of the concrete mixture from the nozzle plays an essential role in the distribution of the components in concrete [72,73].

The spatial distribution of fibers is quantified by orientation angle (φ), as shown in Fig. 23. For printed specimens, the histograms of angle φ were right-skewed compared with the cast specimens, indicating that majority of PP fibers in printed specimens aligned along the printing direction. For the specimens with 0.1% addition of PP fiber, the histogram of angle φ of the printed specimens was concentrated in the range of 45°–90°. Increasing the fiber dosage of 0.3% and 0.5%, the histogram of angle φ of the printed specimens was concentrated in the range of 60°–90°. This can be explained by the increased fiber contact and interaction at high fiber dosages. A similar observation is reported by Pham, Luong [74], who find that more than 90% of steel fibers (6 mm) with the dosage of 0.25%–1% of 3DPC align within 60°–90° along the printing direction.

The fiber orientation factor η_φ is calculated and plotted in function of fiber dosage for both printed and cast specimens in Fig. 24. It can be seen that, the higher the fiber volume dosages, the higher the fiber orientation factor, and the more pronounced the fiber alignment. From Fig. 24, both the shrinkage of the printed and cast specimens was inversely proportional to the fiber orientation factor. Such linear correlation demonstrates that fiber alignment within specimen results in higher flexural strength and the mitigated shrinkage of concrete, especially for the printed specimens along the printing direction.

The results presented in Fig. 15 show that the total shrinkage of the printed and cast specimens without fiber is similar. But when PP fiber is added, the shrinkage of the printed specimens is slightly higher than the cast specimens in the same dosage of PP fiber, though the fiber orientation factor of printed specimens is higher than that of the cast. Furthermore, increasing the PP fiber from 0 to 0.5% mitigates the shrinkage significantly (19.6% reduction maximum at 28 d). Hence, the dosage of PP fiber is the primary factor affecting the shrinkage properties. The addition of PP fiber induces a higher pore volume fraction in the range of 5.5 nm–50 nm of printed specimens compared with cast ones, which explains the higher shrinkage of printed specimens compared with the cast ones [70]. However, it is evident that PP fiber is still effective in enhancing the flexural strength and mitigating the shrinkage by arranging along the printing direction, bridging across pores, thus keeping the internal relative humidity, and reducing the moisture loss during drying.

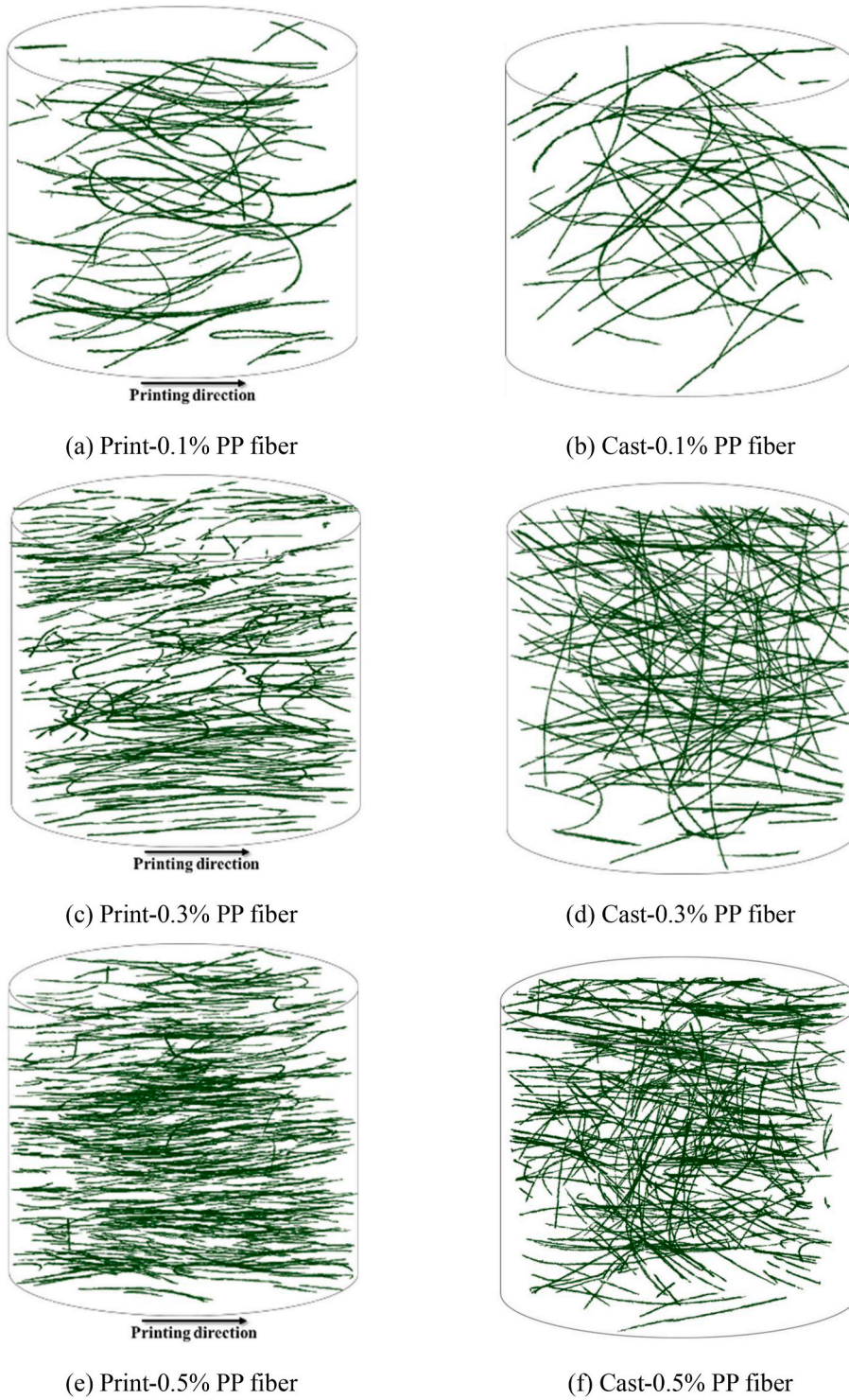
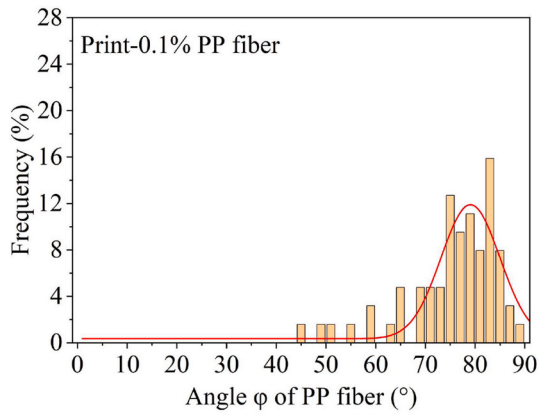
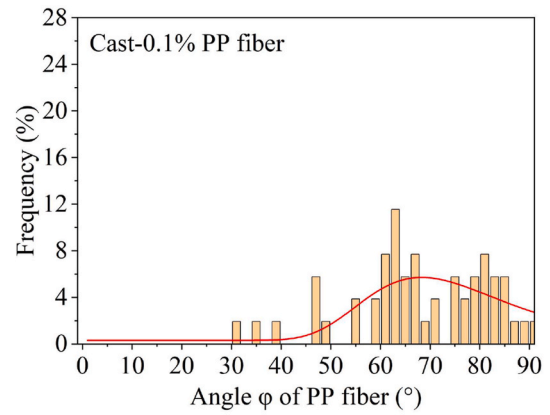


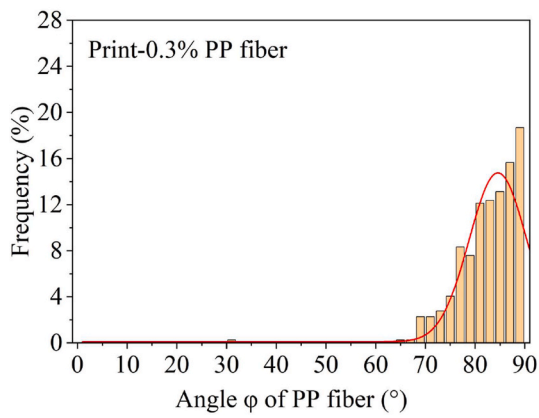
Fig. 22. Fiber orientation of PP fiber with 0.1%–0.5% volume fraction for printed and cast specimens.



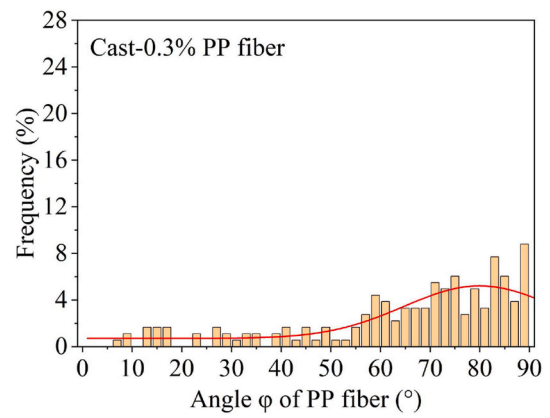
(a)



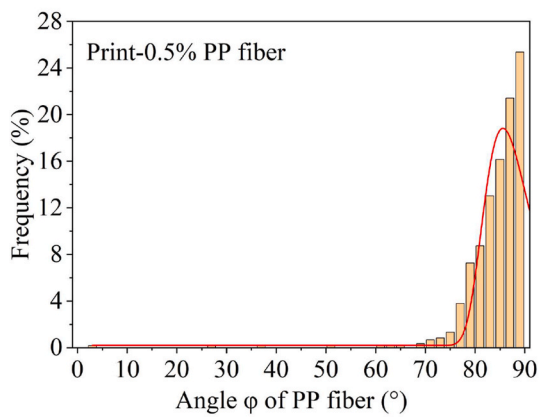
(b)



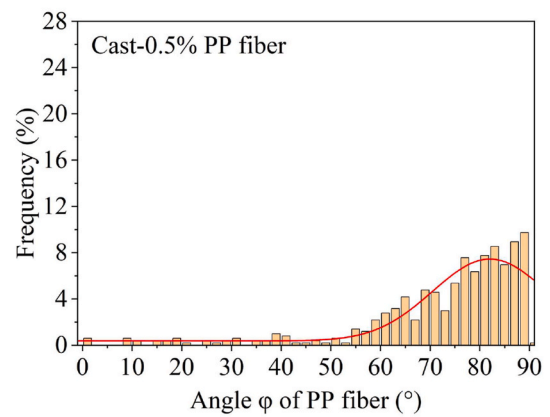
(c)



(d)



(e)



(f)

Fig. 23. Angle ϕ Distributions of PP fiber for printed and cast specimens.

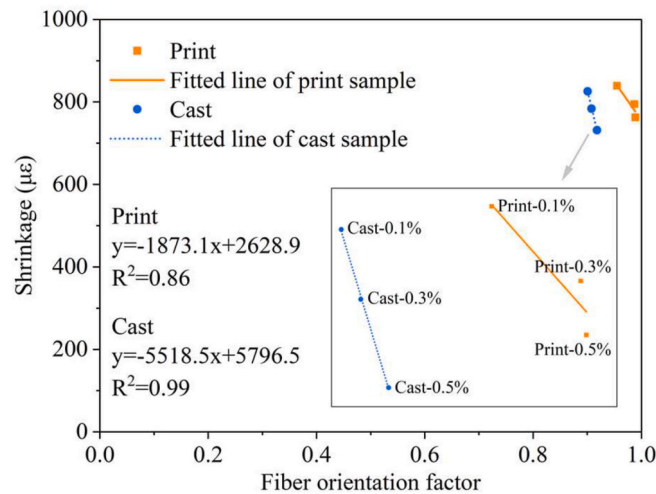


Fig. 24. Shrinkage of printed and cast specimens versus fiber orientation factor.

4. Conclusion

This study investigated the rheological properties, mechanical properties, shrinkage, internal RH, mass loss, pore distribution, and fiber orientation of 3DPC with PP fibers. The influence of fiber incorporation and printing process on the properties of 3DPC was discussed. Main conclusions are summarized as follows:

- (1) Addition of PP fiber increases the static yield stress and plastic viscosity within a certain range, and all the rheologic properties meet the printability of 3DPC.
- (2) Extrusion process forces fibers to align along the printing direction, inducing higher flexural strength of printed specimens measured along the printing direction compared with that of cast specimens. The optimum PP fiber dosage is 0.3% from the mechanical property point.
- (3) The addition of PP fiber increases internal RH and inhibits moisture evaporation. And the mass loss of the specimen is proportional to the total shrinkage.
- (4) The total shrinkage of printed specimen is similar to the cast specimen without fiber. Whereas the shrinkage of printed specimen is higher than the cast one with the addition of fiber. The primary reason is the pore volume fraction in the range of 5.5 nm–50 nm of printed specimen with PP fiber is higher than that of the cast specimen.
- (5) The shrinkage of printed specimen measured along the printing direction is lower than that perpendicularly to the printing direction. This can be attributed to the alignment of fiber along the printing direction. Nonetheless, the shrinkage is influenced, more significantly by the dosage of PP fibers, and less significantly but non-negligibly, by the orientation of fibers.

Although the evolution of total shrinkage properties of printed specimens measured from different directions (with different interfaces) has been investigated and compared with the cast specimens, the effect of surface to volume ratio on 3DPC and shrinkage mitigation strategies by material design considering the low carbon footprint should be investigated further. Specifically, systematically studying on the correlation among shrinkage properties (plastic and drying shrinkage), moisture transport and pore structure of 3DPC under various exposure conditions should be a subject in future research.

Author statement

Lei Ma: Conceptualization, Investigation, Writing- Original Draft, Formal analysis, Data Curation; **Qing Zhang:** Conceptualization, Methodology, Writing - Review & Editing, Supervision, Funding acquisition; **Hélène Lombois-Burger:** Review & Editing, Funding acquisition; **Zijian Jia:** Conceptualization, Methodology; **Zedi Zhang:** Methodology; **Geng Niu:** Methodology; **Yamei Zhang:** Conceptualization, Methodology, Writing - Review & Editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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